DEEP NEURAL NETWORKS

Learning Objectives

- Describe the successive block structure of a deep neural network.
- Build a deep L-layer neural network
- Analyze matrix and vector dimensions to check neural network implementations.
- Explain the role of hyperparameters in deep learning.

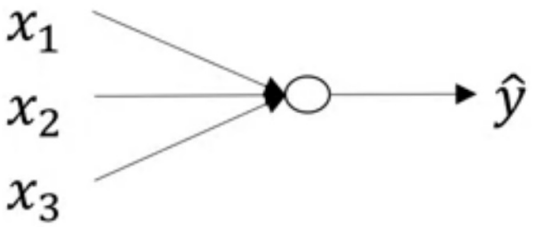


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Deep Neural Networks

Deep L-layer Neural network

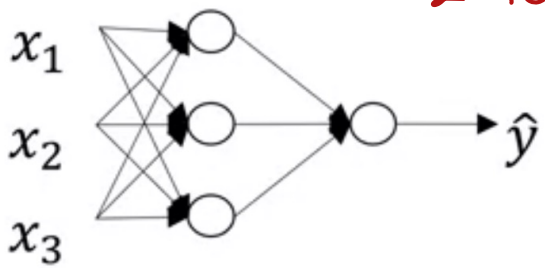
What is a deep neural network?



logistic regression

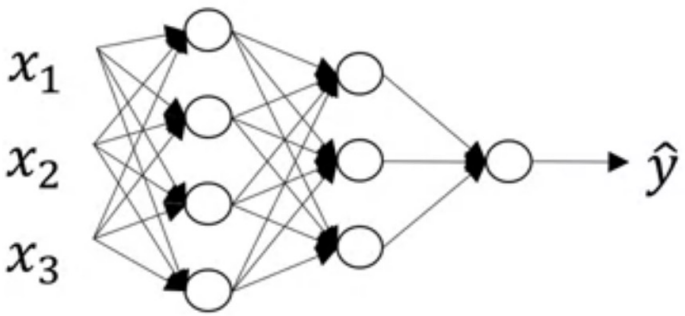
Shallow Model

2-layer NN

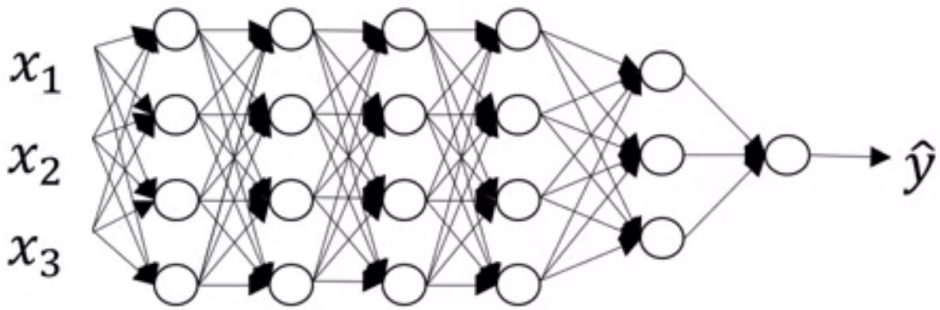


1 hidden layer

"Shallow"



2 hidden layers

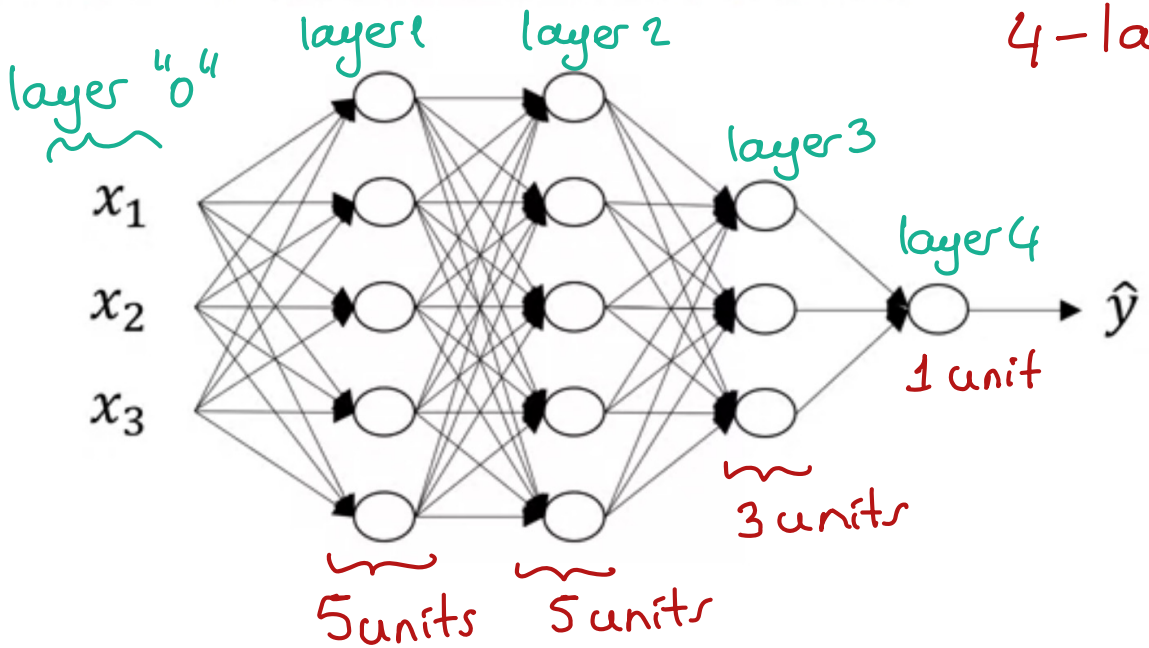


5 hidden layers

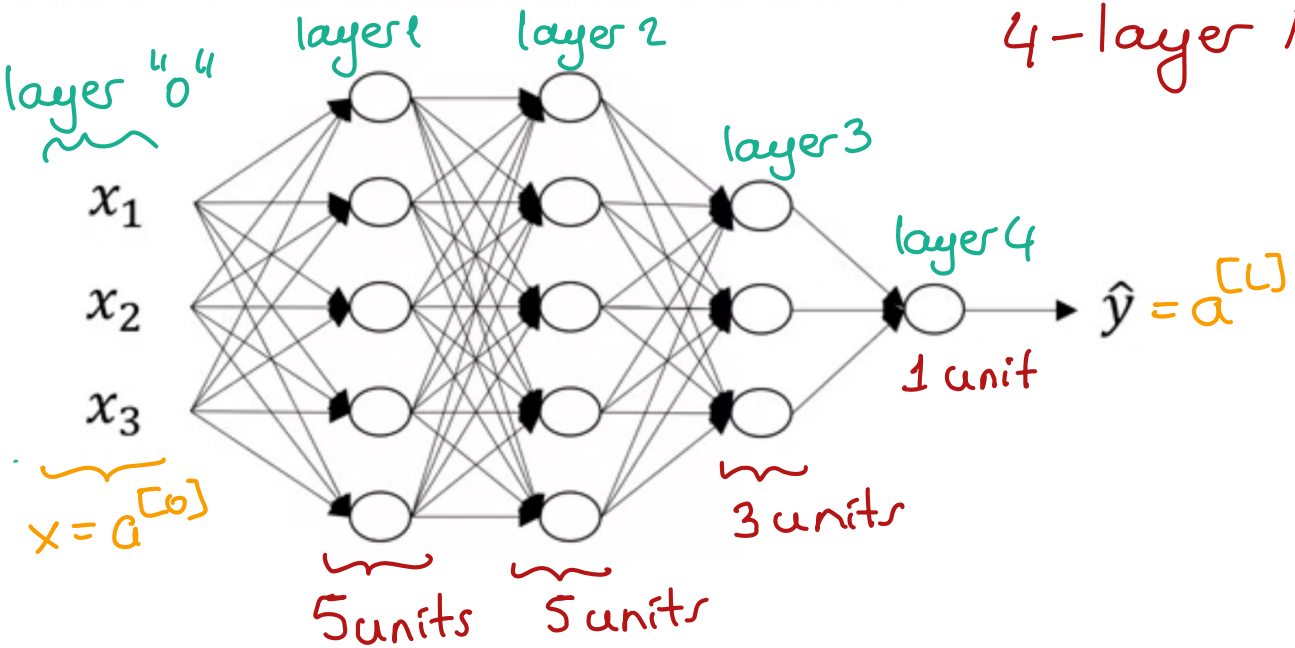
"Deep" model

3-layer NN: do not count the input layer

Deep neural network notation



Deep neural network notation



4-layer NN

$L = 4$ (# layers)

$n^{[l]}$: # units in layer l, $n^{[1]} = 5$, $n^{[2]} = 5$, $n^{[3]} = 3$, $n^{[4]} = n^{[L]} = 1$

$n^{[0]} = n_x = 3$

$a^{[l]}$: activations in layer l, $a^{[0]} = x$ (input), $a^{[L]} = \hat{y}$ (output)

$a^{[l]} = g^{[l]}(z^{[l]})$,

$W^{[l]}$: weights for computing $z^{[l]}$

$b^{[l]}$: bias terms for layer $[l]$

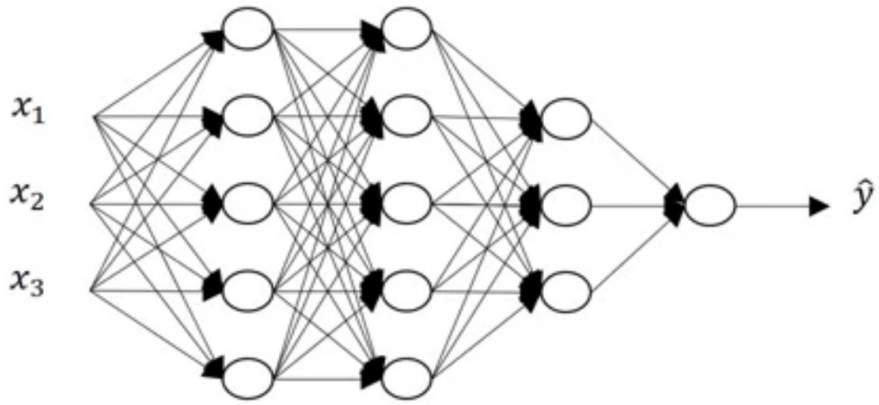


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Deep Neural Networks

Forward Propagation in a Deep Network

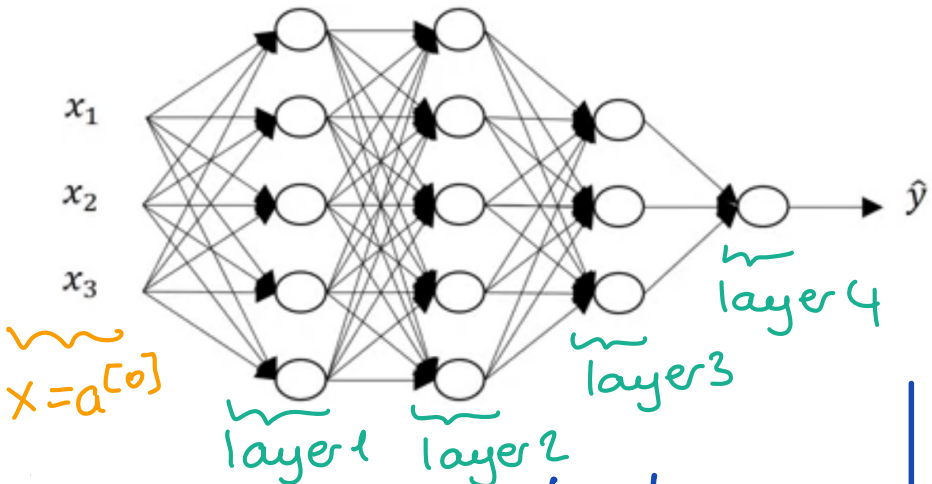
Forward propagation in a deep network



Goals:

1) Study forward propagation for a single training example x
2) Carry out forward propagation on the entire training set at the same time (using vectorization).

Forward propagation in a deep network



Forward propagation for a single example x

Compute layer 1 activations:

$$z^{[1]} = W^{[1]} a^{[0]} + b^{[1]}$$

$$a^{[1]} = g^{[1]}(z^{[1]})$$

Compute layer 2 activations:

$$z^{[2]} = W^{[2]} a^{[1]} + b^{[2]}$$

$$a^{[2]} = g^{[2]}(z^{[2]})$$

Compute layer 3 activations:

$$z^{[3]} = W^{[3]} a^{[2]} + b^{[3]}$$

$$a^{[3]} = g^{[3]}(z^{[3]})$$

Compute layer 4 activations:

$$z^{[4]} = W^{[4]} a^{[3]} + b^{[3]}$$

$$a^{[4]} = g^{[4]}(z^{[4]}) = \hat{y} \text{ (output)}$$

- The equations of all 4 layers look the same.
  General rule for forward propagation equation:

$$z^{[l]} = W^{[l]} a^{[l-1]} + b^{[l]}$$
$$a^{[l]} = g^{[l]}(z^{[l]})$$

$a^{[l]}$: activations of layer l; $g^{[l]}$: activation functions of layer l

- Forward Propagation for the whole training set at the same time:

Vectorized:

layer 1
$$Z^{[1]} = W^{[1]} X^{[1]} + b^{[1]}$$
$$A^{[1]} = g^{[1]}(Z^{[1]})$$

layer 2
$$Z^{[2]} = W^{[2]} A^{[1]} + b^{[2]}$$
$$A^{[2]} = g^{[2]}(Z^{[1]})$$

layer 3 { ⋮

layer 4
$$Z^{[4]} = W^{[4]} A^{[3]} + b^{[4]}$$
$$A^{[4]} = g^{[4]}(Z^{[4]}) = \hat{Y}$$

(X contains all samples in its columns)

$$X^{[1]} = A^{[0]}$$

Similarly e.g.:

$$Z^{[2]} = \begin{bmatrix} | & | & & | \\ z^{[2](1)} & z^{[2](2)} & \cdots & z^{[2](m)} \\ | & | & & | \end{bmatrix}$$

z of one sample in its column

$$\hat{Y} = \begin{bmatrix} \hat{y}^{(1)} & \hat{y}^{(2)} & \cdots & \hat{y}^{(m)} \end{bmatrix}$$

- We replaced lowercase letters with uppercase counterparts.

$$z^{[l]} = W^{[l]} a^{[l-1]} + b^{[l]}$$
$$a^{[l]} = g^{[l]}(z^{[l]})$$

single example

$z \leftarrow Z$
$a \leftarrow A$

$$Z^{[l]} = W^{[l]} A^{[l-1]} + b^{[l]}$$
$$A^{[l]} = g^{[l]}(Z^{[l]})$$

all examples of training set

- When implementing a neural network you need a for loop to activations of all layers one after another. (No way to eliminate this loop.)

- When implementing a deep neural network, the best way to debug the code is to check the dimensions of the matrices and vectors.

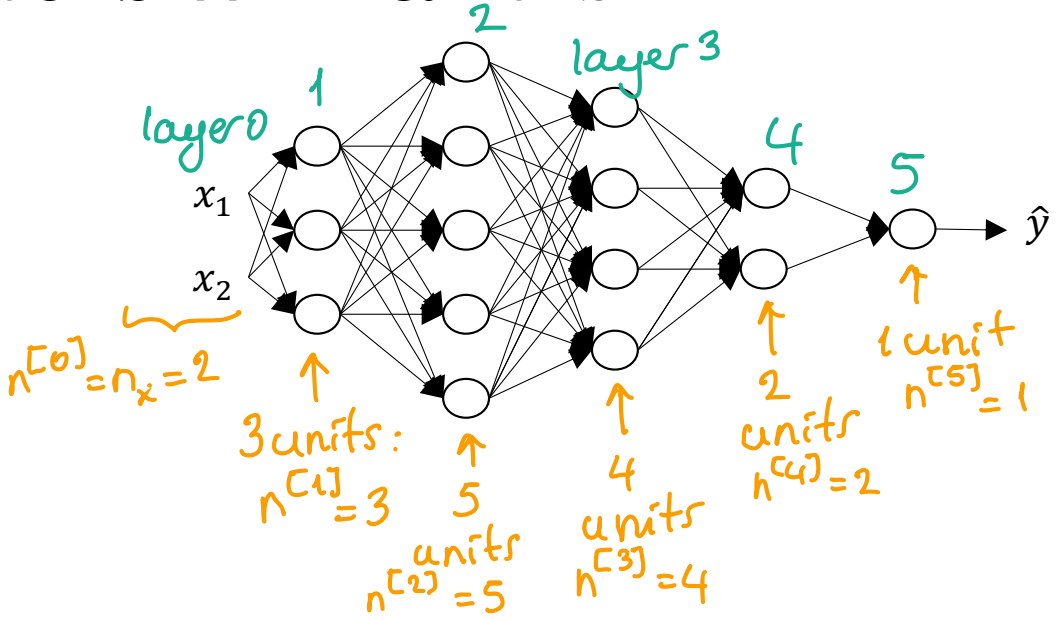


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Deep Neural Networks

Getting your matrix dimensions right

Parameters $W^{[l]}$ and $b^{[l]}$



$L=5$ # layers

$n^{[0]}=n_x=2$

$n^{[1]}=3$ # units in layer 1

$n^{[2]}=5$

$n^{[3]}=4$

$n^{[4]}=2$

$n^{[5]}=1$

Forward propagation:

$$z^{[1]} = W^{[1]} x + b^{[1]}$$

Dimens: $(3,1)$ $(3,2)$ $(2,1)$ $(3,1)$

$(n^{[1]},1)$ $(n^{[1]},n^{[0]})$ $(n^{[0]},1)$ $(n^{[1]},1)$

$$\begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix} = \begin{bmatrix} \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \end{bmatrix} \begin{bmatrix} \cdot \\ \cdot \end{bmatrix}$$

more generally $\Rightarrow$

$W^{[1]}: (n^{[1]}, n^{[0]})$

$W^{[2]}: (n^{[2]}, n^{[1]}) \rightarrow (5,3)$

$$z^{[2]} = W^{[2]} a^{[1]} + b^{[2]}$$

$(5,1)$ $(5,3)$ $(3,1)$ $(5,1): (n^{[2]},1)$

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↓similarly

$W^{[3]} : (4,5)$

$W^{[4]} : (2,4)$

$W^{[5]} : (1,2)$

More generally:

$W^{[l]} : (n^{[l]}, n^{[l-1]})$

$b^{[l]} : (n^{[l]}, 1)$ (same size as $z^{[l]}$)

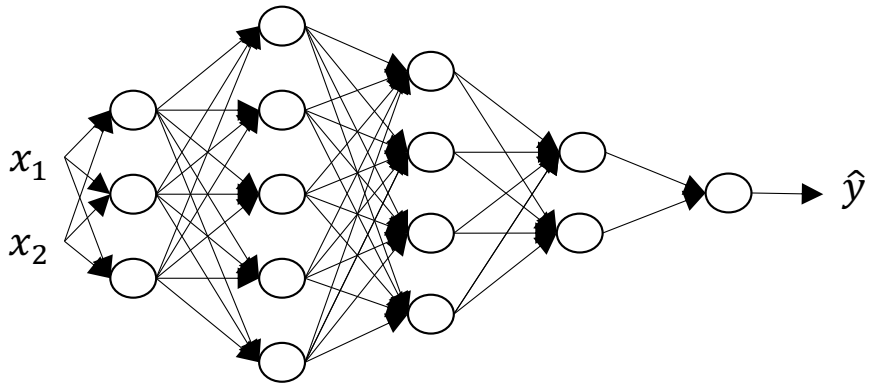
$dW^{[l]} : (n^{[l]}, n^{[l-1]})$ (same dimens. as $W^{[l]}$)

$db^{[l]} : (n^{[l]}, 1)$ (same dimens. as $b^{[l]}$)

$a^{[l]} = g^{[l]}(z^{[l]})$

$(n^{[l]}, 1)$

Vectorized implementation (for all training data)



- Dimensions of w, b, dw, db will stay the same.
- Dimensions of z, a, x will change.

Previously: $z^{[1]} = w^{[1]} x + b^{[1]}$

$(n^{[1]},1)\ (n^{[1]},n^{[0]})\ (n^{[0]},1)\ (n^{[1]},1)$

Vectorized: $Z^{[1]} = w^{[1]} X + b^{[1]}$

$(n^{[1]},m)\ (n^{[1]},n^{[0]})\ (n^{[0]},m)\ (n^{[1]},1)$

$$Z^{[1]} = \begin{bmatrix} | & | & & | \\ z^{[1](1)} & z^{[1](2)} & \cdots & z^{[1](m)} \\ | & | & & | \end{bmatrix}$$

m: size of training set

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will be broadcasted for addition with $(n^{[l]}, m)$ matrix

$(n^{[l]}, m)$

In general:

$$z^{[l]}, a^{[l]} : (n^{[l]}, 1)$$

$$Z^{[l]}, A^{[l]} : (n^{[l]}, m)$$

if $l=0 \Rightarrow A^{[0]} = X = (n^{[0]}, m)$

backprop: $dZ^{[l]}, dA^{[l]} : (n^{[l]}, m)$



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Deep Neural Networks

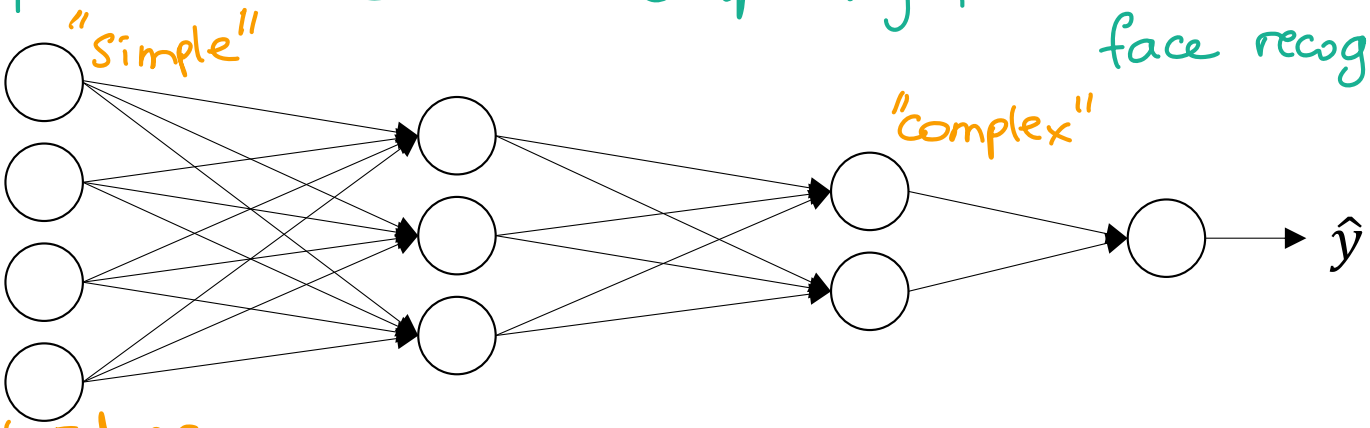
Why deep representations?

Why might deep networks work well?

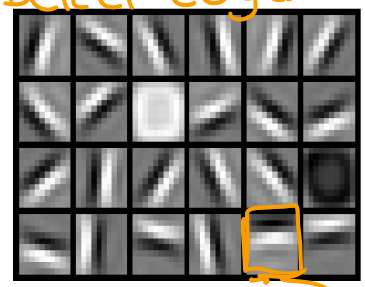
Intuition about deep representation

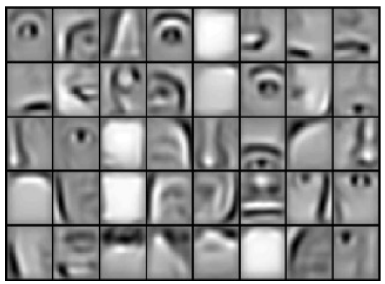
What is a deep neural network computing for face detection or face recognition?





Detect Edges







(will make more sense after CNNs)

4x6 = 24 hidden units visualized by each square box.

this neuron detects horizontal edges

group edges to form parts of faces.

put face parts together to detect different types of faces.

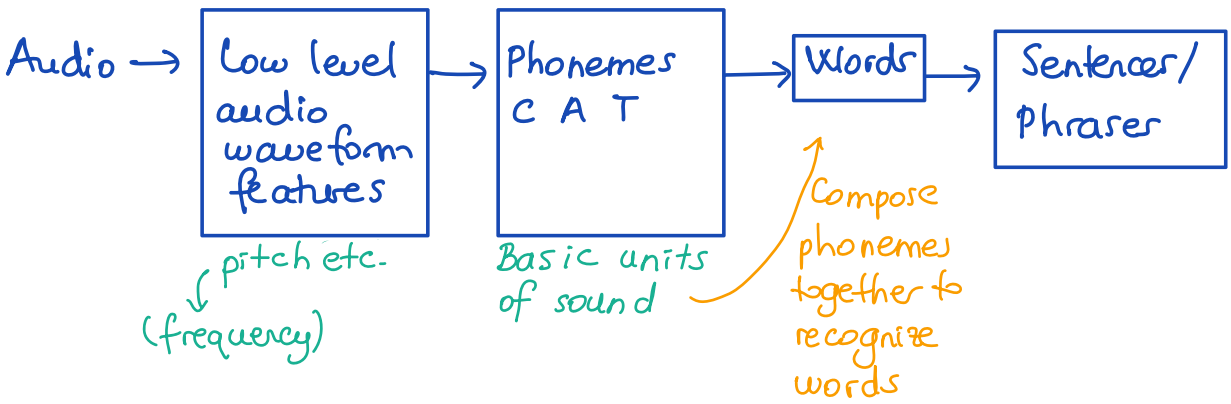
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- Earlier layers of a NN detect "simple" functions, like edges. Then, compose them together in the later layers of a neural network, to learn more complex functions.

- Edge detectors are looking at smaller parts of an image, whereas facial detectors look at a larger areas of an image.

- Simple to complex hierarchical data representation.

Other application : Speech recognition



Circuit theory and deep learning

Informally: There are functions you can compute with a "small" L-layer deep neural network that shallower networks require exponentially more hidden units to compute.

small # of hidden units.

- There are mathematical functions, that are much easier to compute with deep networks than with shallow networks.



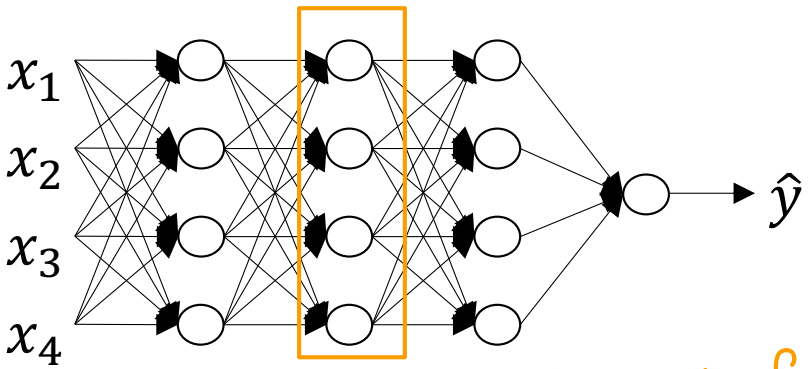
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Deep Neural Networks

Building blocks of deep neural networks

Goal: Put forward and backpropagation components together to build a deep network.

Forward and backward functions



Pick layer l and focus on computations of layer l.

Parameters: $W^{[l]}, b^{[l]}$

Forward Propagat.: Input: $a^{[l-1]}$, Output: $a^{[l]}$

$z^{[l]} = W^{[l]} a^{[l-1]} + b^{[l]}$: cache $z^{[l]}$ for backpropagation

$a^{[l]} = g^{[l]}(z^{[l]})$

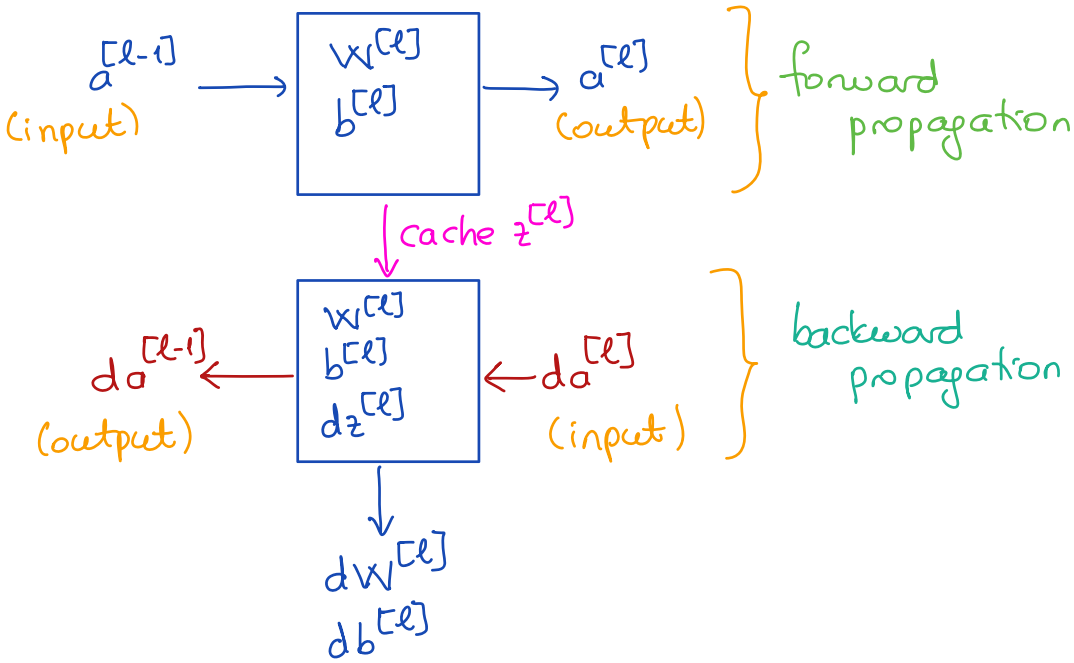
Backprop: Input $da^{[l]}$, cache $z^{[l]}$

output: $da^{[l-1]}, dW^{[l]}, db^{[l]}$

for gradient descent

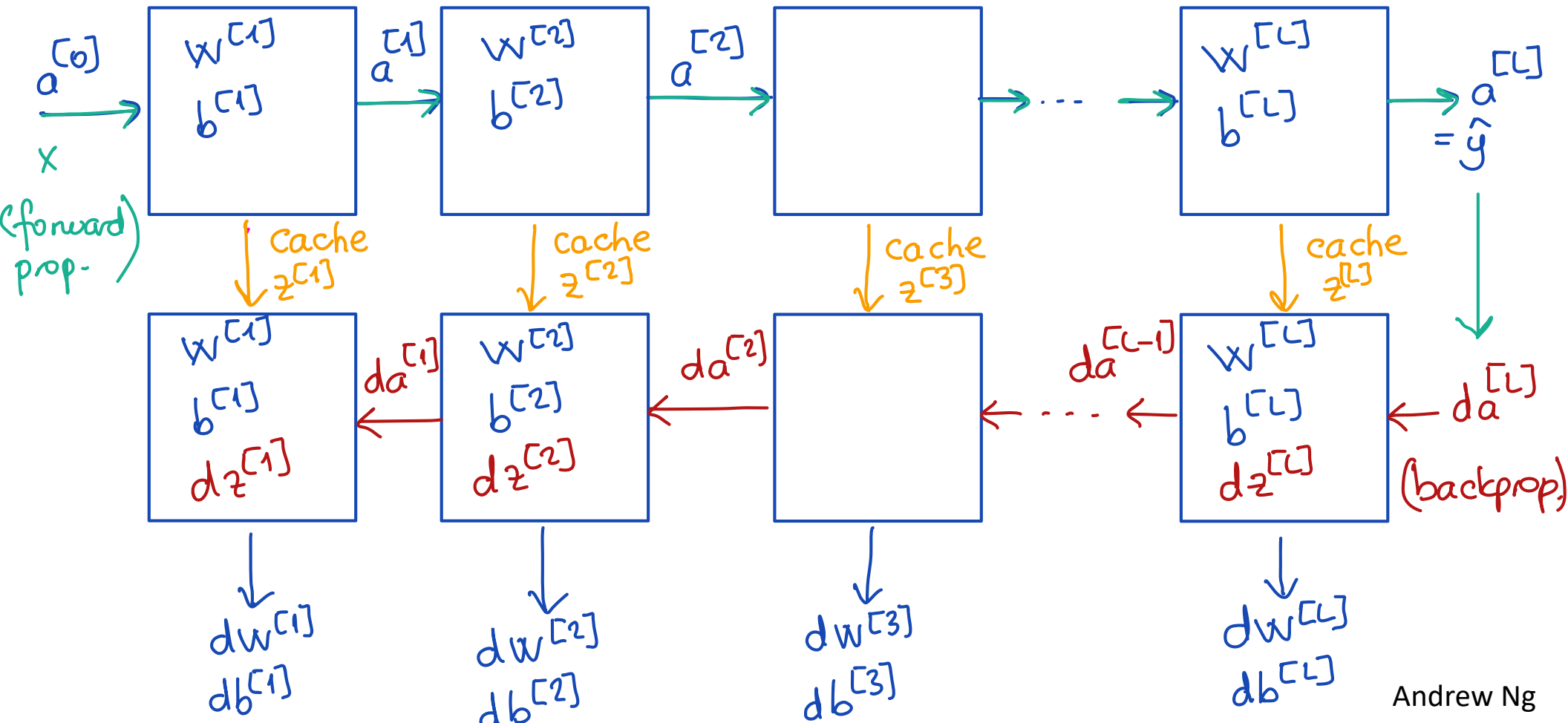
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SUMMARY for layer l:



Forward and backward functions:

One iteration of gradient descent



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update parameters: $W^{[l]} := W^{[l]} - \alpha dW^{[l]}$, $b^{[l]} := b^{[l]} - \alpha db^{[l]}$



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Deep Neural Networks

Forward and backward propagation

Goal: Implement the forward and backward propagation blocks given above.

Forward propagation for layer l

Input $a^{[l-1]}$

Output $a^{[l]}$, cache $(z^{[l]})$

$$z^{[l]} = W^{[l]} \cdot a^{[l-1]} + b^{[l]}$$

$$a^{[l]} = g^{[l]}(z^{[l]})$$

First input:

$a^{[0]} = x$: first sample

Vectorized:

$$Z^{[l]} = W^{[l]} \cdot A^{[l-1]} + b^{[l]}$$

$$A^{[l]} = g^{[l]}(Z^{[l]})$$

First input

$A^{[0]} = X$ = matrix of whole training set

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Backward propagation for layer l

Input $da^{[l]}$

Output $da^{[l-1]}, dW^{[l]}, db^{[l]}$

$$dz^{[l]} = da^{[l]} * g^{[l]\prime}(z^{[l]})$$

$$dW^{[l]} = dz^{[l]} \cdot a^{[l-1]T}$$

$$db^{[l]} = dz^{[l]}$$

$$da^{[l-1]} = W^{[l]T} dz^{[l]}$$

insert here ⇒

$$dz^{[l]} = W^{[l+1]T} dz^{[l+1]} * g^{[l]\prime}(z^{[l]})$$

$*$: element-wise product

Vectorized version:

$$dZ^{[l]} = dA^{[l]} * g^{[l]\prime}(Z^{[l]})$$

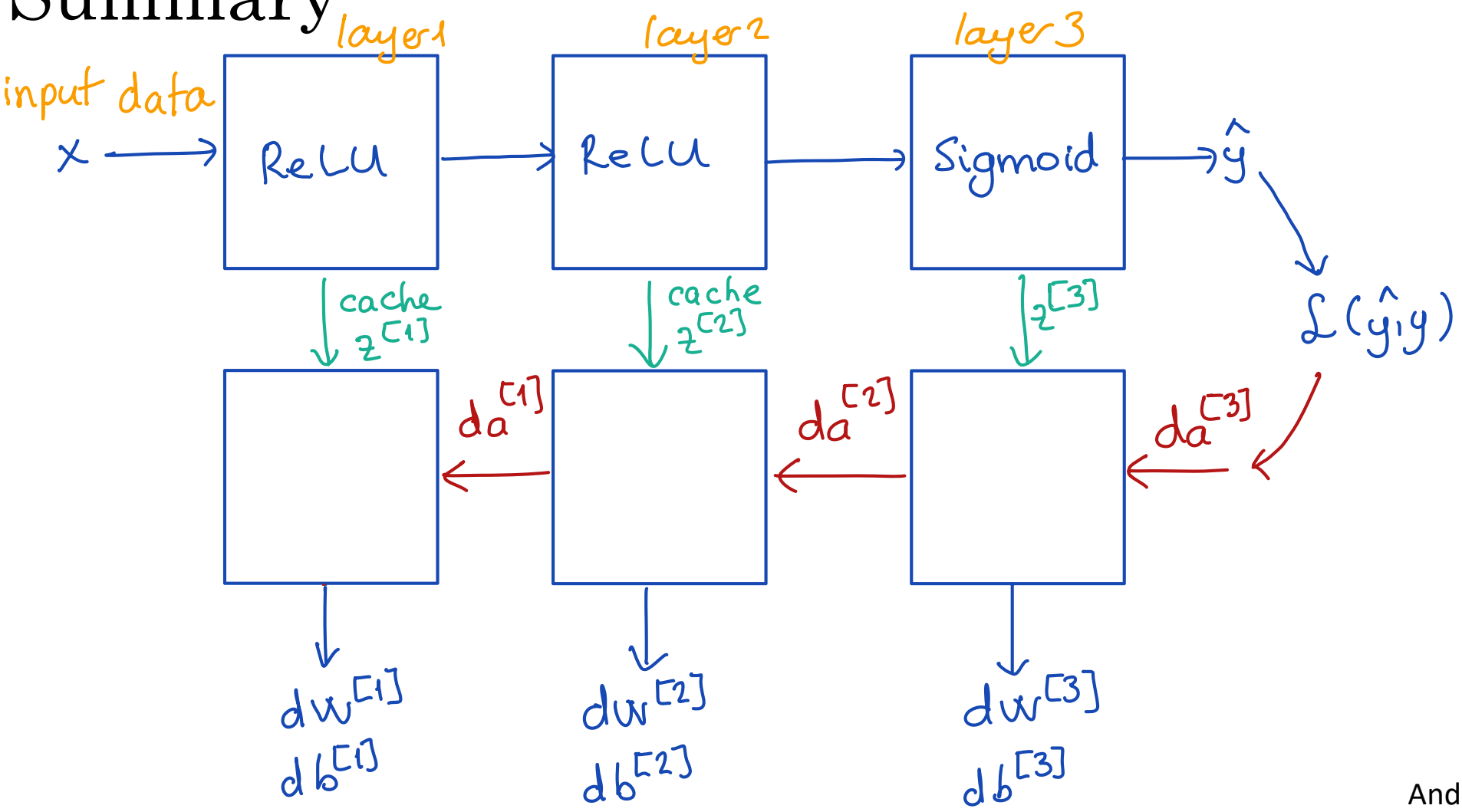
$$dW^{[l]} = \frac{1}{m} dZ^{[l]} \cdot A^{[l-1]T}$$

$$db^{[l]} = \frac{1}{m} \text{np.sum}(dZ^{[l]}, \text{axis}=1, \text{keepdims}=\text{True})$$

$$dA^{[l-1]} = W^{[l]T} \cdot dZ^{[l]}$$

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Summary



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NOTES

- When using sigmoid activation at the last layer L:
  Loss is the binary cross entropy loss. Then it can be shown that:

$$da^{[L]} = \frac{d\mathcal{L}}{da} = -\frac{y}{a} + \frac{1-y}{1-a}$$

↓

we initialize the backward recursion with this

in vectorized version: we initialize backward recursion as follows:

$$dA^{[L]} = \left(-\frac{y^{(1)}}{a^{(1)}} + \frac{(1-y^{(1)})}{(1-a^{(1)})} + \dots + -\frac{y^{(m)}}{a^{(m)}} + \frac{(1-y^{(m)})}{(1-a^{(m)})} \right)$$



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Deep Neural Networks

Parameters vs Hyperparameters

What are hyperparameters?

Parameters: $W^{[1]}$, $b^{[1]}$, $W^{[2]}$, $b^{[2]}$, $W^{[3]}$, $b^{[3]}$ ...

Hyperparameters:

- Learning rate α
- # iterations
- # hidden layers
- # hidden units $n^{[1]}, n^{[2]}, \cdots$
- choice of activation function

Parameters that control the ultimate parameters W, b $\Rightarrow$ hyperparameters

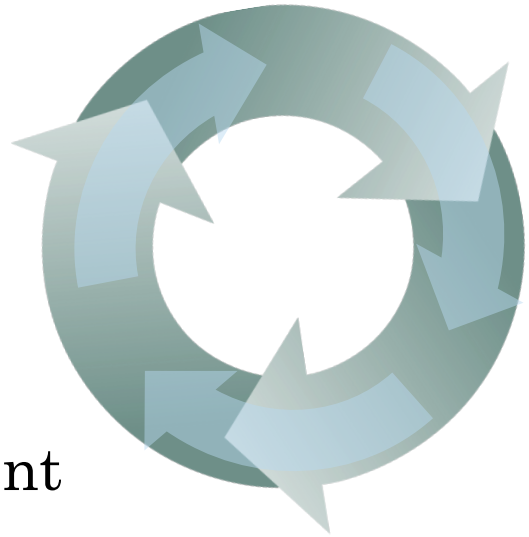
Later: Momentum, minibatch size, regularizations, ---

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Applied deep learning is a very empirical process

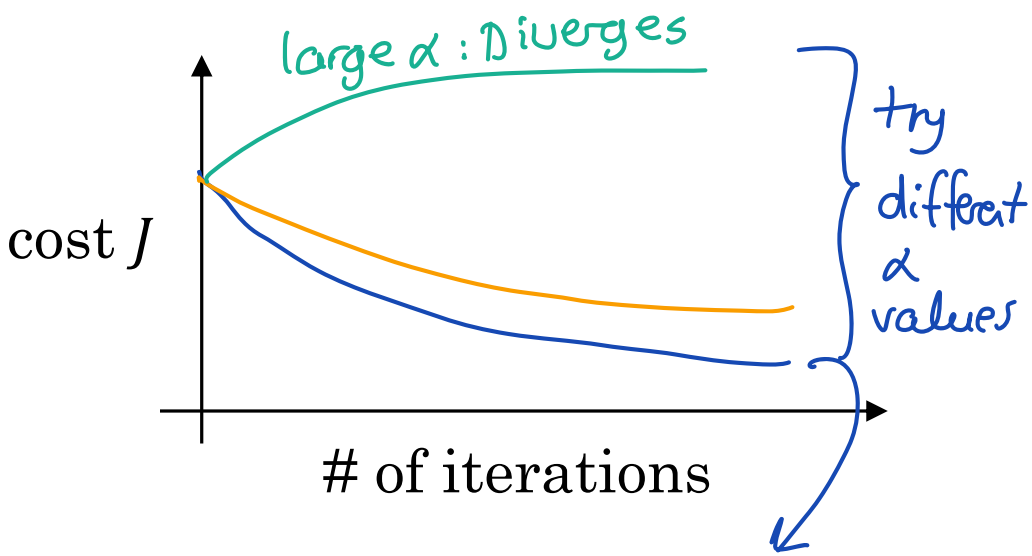
Idea

$\alpha = 0.04$ (learning rate)



Experiment

Code



- Try many hyperparameters and choose the one that works best.

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Deep Neural Networks

What does this have to do with the brain?

Forward and backward propagation

Backward Prop.

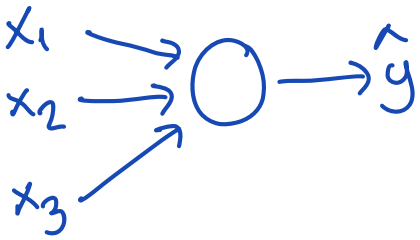
Forward Prop.

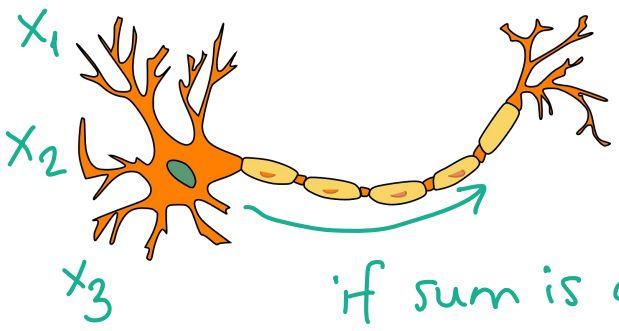
$$Z^{[1]} = W^{[1]}X + b^{[1]}$$
$$A^{[1]} = g^{[1]}(Z^{[1]})$$
$$Z^{[2]} = W^{[2]}A^{[1]} + b^{[2]}$$
$$A^{[2]} = g^{[2]}(Z^{[2]})$$
$$\vdots$$
$$A^{[L]} = g^{[L]}(Z^{[L]}) = \hat{Y}$$

$$dZ^{[L]} = A^{[L]} - Y$$
$$dW^{[L]} = \frac{1}{m} dZ^{[L]} A^{[L-1]T}$$
$$db^{[L]} = \frac{1}{m} np.\text{sum}(dZ^{[L]}, axis = 1, keepdims = True)$$
$$dZ^{[L-1]} = dW^{[L]T} dZ^{[L]} * g'^{[L-1]}(Z^{[L-1]})$$
$$\vdots$$
$$dZ^{[1]} = dW^{[2]T} dZ^{[2]} * g'^{[1]}(Z^{[1]})$$
$$dW^{[1]} = \frac{1}{m} dZ^{[1]} A^{[0]T} \qquad (A^{[0]T} = X^T)$$
$$db^{[1]} = \frac{1}{m} np.\text{sum}(dZ^{[1]}, axis = 1, keepdims = True)$$

"It's like the brain."

Overly simplified





'if sum is above a threshold a signal is sent.

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